

AN INTERNSHIP REPORT
ON
FULL STACK DEVELOPMENT

Submitted to

School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh

In partial fulfilment of the requirements for the degree of
Bachelor of Technology (Computer Science and Engineering)



Submitted By

DAKSH PRATAP SINGH

(Roll No. 2410030168)

B.Tech (Computer Science and Engineering)

Academic Year 2026

CANDIDATE'S DECLARATION

I, Daksh Pratap Singh, hereby declare that the internship report entitled “Full Stack Development Internship at Thiranex” has been prepared by me as part of the academic requirements for the Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

This report presents the work, projects, learning outcomes, and practical exposure gained during my internship from 20 August 2026 to 19 September 2026. The project descriptions and internship details have been prepared from the tasks assigned to me and the official internship documents provided by Thiranex.

I further declare that this report has been prepared for academic submission and that the work described represents my internship learning and project activities.

Student Name: Daksh Pratap Singh | Roll No.: 2410030168

Intern ID: THX-AUG2026-405

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Thiranex for providing me the opportunity to undertake an internship in Full Stack Development. The internship provided a practical environment in which I could apply concepts related to web development, frontend design, backend APIs, databases, authentication, and full-stack application development.

I am thankful to the mentors and the internship team for providing project-based tasks and guidance throughout the internship. Working on multiple applications helped me understand how individual web technologies work together to form a complete software application.

I also express my gratitude to the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic support and encouragement. Finally, I am thankful to my parents and everyone who supported me during the internship period.

Student Name: Daksh Pratap Singh | Roll No.: 2410030168

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1. INTERNSHIP COMPLETION CERTIFICATE

The official internship completion certificate is reproduced below.



The certificate confirms that Daksh Pratap Singh successfully completed an internship in Full Stack Development from 20 August 2026 to 19 September 2026.

2. INTERNSHIP OFFER LETTER

The official offer letter is reproduced below.



Date: 20 August 2026

TO,

Daksh Pratap Singh

Intern ID: THX-AUG2026-405

Subject: Offer Letter for Internship in Full Stack Development

Dear **Daksh Pratap Singh**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - Full Stack Development
Commencement Date	20 Aug 2026
Completion Date	19 Sept 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



The offer letter identifies the role as Intern – Full Stack Development, with a remote/project-based work mode and an internship period from 20 August 2026 to 19 September 2026.

3. PROJECT DESCRIPTION

3.1 Introduction

This report presents the practical learning and project work completed during my Full Stack Development internship at Thiranex. The internship was conducted from 20 August 2026 to 19 September 2026 in a remote and project-based mode. The official offer letter states that the internship involved practical projects under industry mentorship and periodic progress review.

The internship focused on understanding the complete lifecycle of web application development. The assigned tasks progressed from building a personal portfolio to developing applications involving task management, e-commerce functionality, and blogging with comments. These projects provided exposure to frontend development, backend/API development, database integration, authentication, authorization, CRUD operations, responsive design, and application deployment concepts.

3.2 Organization and Internship Profile

Thiranex provided the internship opportunity in the role of Intern – Full Stack Development. According to the official offer letter, the internship was remote/project-based and involved practical projects under industry mentorship. The internship commenced on 20 August 2026 and was completed on 19 September 2026.

Particular	Details
Organization	Thiranex
Role	Intern – Full Stack Development
Intern ID	THX-AUG2026-405
Start Date	20 August 2026
Completion Date	19 September 2026
Work Mode	Remote / Project-Based
Duration	Approximately 4 weeks

3.3 Problem Statement

The internship addressed the practical gap between learning individual web-development technologies and using them together to build complete applications. The project tasks required developing applications with user-facing interfaces, backend services, data storage, authentication, and application-level features. The objective was therefore to gain hands-on experience in designing and implementing functional full-stack web applications rather than working only with isolated programming concepts.

3.4 Objectives of the Internship

- To understand the fundamentals of full-stack web application development.

- To strengthen frontend development skills using HTML, CSS, JavaScript and related frameworks where applicable.
- To understand backend development and REST-style API integration.
- To learn how application data can be stored and managed using relational or NoSQL databases.
- To implement authentication and authorization concepts in web applications.
- To practice CRUD operations for real-world application features.
- To develop responsive interfaces suitable for web and mobile screens.
- To gain practical experience by completing multiple project-based assignments.
- To understand the basic deployment workflow for a live web application.

3.5 Scope of the Internship

The scope of the internship covered the major layers of a modern full-stack application. The assigned projects included frontend interfaces, backend APIs, database integration, authentication, authorization, product and order management, task management, blogging features, comments, and responsive design. The portfolio project also included deployment of the website on a hosting platform as part of the task requirements.

The internship was project-based. The task dashboard provided four major development assignments: Personal Portfolio Website, Task Management Application, E-Commerce Web Application, and Blog Platform with Comments. Each assignment focused on a different set of full-stack development requirements.

3.6 Technologies and Tools Used

Area	Technologies / Concepts
Frontend	HTML, CSS, JavaScript; React.js where selected
Backend	Node.js / Express.js or Django / Flask, according to project implementation
Database	MySQL, PostgreSQL or MongoDB
APIs	RESTful APIs / backend API integration
Authentication	User login, registration, authentication and authorization
Application Features	CRUD operations, cart, checkout, task tracking, comments
Responsive Design	Responsive layouts for web and mobile screens
Deployment	Hosting platforms such as Heroku, Netlify or Vercel
Development Practice	Project-based development, testing, debugging and iteration

3.7 Completed Internship Projects

All four assigned internship projects were successfully completed and submitted on track. The completed-work dashboard screenshots below provide supporting evidence of the project submissions.

3.7.1 Completed Work – Personal Portfolio Website

The first project was a full-stack personal portfolio website designed to showcase projects, technical skills and personal information. The task required a frontend built using HTML, CSS and JavaScript or React.js, a backend using Node.js/Express.js or Django/Flask, and database storage using MySQL, MongoDB or PostgreSQL. The project also included deployment on a hosting platform.

- Designed a structured portfolio interface to present projects and skills.
- Worked with frontend and backend components as parts of one application.
- Understood the role of a database in storing project-related information.
- Learned the basic workflow of deploying a web application.

3.7.2 Completed Work – Task Management Application

The second project was a task management web application for creating, updating and tracking tasks. The task requirements included user authentication and authorization, CRUD operations, responsive design, and optional real-time updates through WebSockets.

- Implemented the concept of user authentication and access control.
- Worked with task creation, updating, deletion and tracking.
- Understood CRUD operations as a practical application pattern.
- Considered responsive design for different screen sizes.
- Learned how real-time updates can be added to a web application.

3.7.3 Completed Work – E-Commerce Web Application

The third project was a basic e-commerce web application. Its required features included a product catalogue, add-to-cart and checkout functionality, user login, role-based access for Admin/User, backend APIs for product and order management, and database integration.

- Worked with product listing and catalogue concepts.
- Understood the flow of adding products to a shopping cart.
- Studied the relationship between users, products and orders.
- Applied authentication and role-based access concepts.
- Understood how backend APIs support product and order management.

3.7.4 Completed Work – Blog Platform with Comments

The fourth project was a blogging platform in which users can create posts and interact through comments. The task included user registration, login and authentication, CRUD operations for blog posts, a comment section, RESTful APIs and database integration.

- Implemented or practiced user registration and login workflows.
- Worked with create, edit and delete operations for blog posts.
- Understood user interaction through comments.
- Studied RESTful backend API design and database integration.
- Connected content-management features with user authentication.

3.7.5 Internship Task Dashboard

The internship task dashboard listed the four project assignments and their key requirements. The screenshots below are included as supporting evidence of the assigned project work.

The dashboard displays four completed tasks, each with a numbered header, status, due date, submission status, title, description, tutorial link, key features, expected outcome, and a completion confirmation.

Task 1: Personal Portfolio Website
Status: COMPLETED
Due Date: 27 Aug 2026
Submitted on track
Description: Build a full-stack personal portfolio to showcase your projects and skills.
Tutorial: Watch Task Tutorial
Key Features:

- Frontend: HTML, CSS, JavaScript (or React.js)
- Backend: Node.js/Express.js or Django/Flask
- Database: Store project details using MySQL, MongoDB, or PostgreSQL
- Deploy website on a hosting platform (Heroku, Netlify, or Vercel)

Expected Outcome: Gain experience in integrating frontend, backend, and database for a live project.
Completed Successfully!

Task 2: Task Management Application
Status: COMPLETED
Due Date: 03 Sept 2026
Submitted on track
Description: Develop a task management web application for creating, updating, and tracking tasks.
Tutorial: Watch Task Tutorial
Key Features:

- User authentication & authorization
- CRUD operations for tasks
- Real-time updates (optional: using WebSockets)
- Responsive design for web and mobile screens

Expected Outcome: Learn full-stack application structure, API integration, and dynamic data handling.
Completed Successfully!

Task 3: E-Commerce Web Application
Status: COMPLETED
Due Date: 10 Sept 2026
Submitted on track
Description: Build a basic online store with product management and order tracking.
Tutorial: Watch Task Tutorial
Key Features:

- Product catalog, add to cart, and checkout functionality
- User login and role-based access (Admin/User)
- Backend APIs for product & order management
- Database integration with MySQL, PostgreSQL, or MongoDB

Expected Outcome: Gain hands-on experience in building a complex full-stack application with real-world features.
Completed Successfully!

Task 4: Blog Platform with Comments
Status: COMPLETED
Due Date: 17 Sept 2026
Submitted on track
Description: Develop a blogging platform where users can create posts and comment.
Tutorial: Watch Task Tutorial
Key Features:

- User registration, login, and authentication
- Create, edit, delete blog posts
- Comment section for user interaction
- Backend with RESTful APIs and database integration

Expected Outcome: Learn full-stack development with content management features and user interaction.
Completed Successfully!

3.8 System Architecture and Application Flow

The projects followed a general full-stack architecture in which the user interacts with the frontend, the frontend communicates with backend APIs, and the backend handles business logic and database operations. Authentication and authorization are applied where required by the application.

Layer	Responsibility
User / Client Layer	User interaction through web pages and responsive interfaces.
Frontend Layer	UI, forms, navigation, validation and communication with APIs.
API / Backend Layer	Business logic, request handling, authentication and CRUD operations.
Database Layer	Persistent storage of users, tasks, products, orders, posts and comments.
Deployment Layer	Hosting and serving the completed web application.

3.9 Methodology and Work Process

The internship used a project-based methodology. Each assignment began with studying the requirements and identifying the application components. The work was then divided into frontend, backend and data-management activities. Features were implemented incrementally and tested during development.

Phase	Activities
1. Requirement Understanding	Read project requirements and identify required features.
2. Planning	Decide application structure, pages, data entities and API requirements.
3. Frontend Development	Create layouts, forms, navigation and responsive interfaces.
4. Backend Development	Develop server-side logic and REST-style API endpoints.
5. Database Integration	Connect application features with persistent data storage.
6. Authentication	Implement login/registration and access-control concepts where required.
7. Testing & Debugging	Check application behavior, identify errors and refine features.
8. Deployment	Understand and apply the basic process of hosting a completed application.

3.10 Learning Outcomes

- Improved understanding of the complete frontend-backend-database relationship.
- Practical exposure to building responsive web interfaces.
- Understanding of backend APIs and how frontend applications consume them.
- Practical understanding of CRUD operations.
- Understanding of authentication and authorization in web applications.
- Exposure to database integration using SQL or NoSQL database systems.
- Understanding of application features such as carts, orders, task tracking, posts and comments.
- Improved debugging, problem-solving and project-organization skills.
- Basic understanding of web application deployment and hosting.
- Greater confidence in approaching a full-stack application from requirements to implementation.

3.11 Challenges and Skills Developed

Working on several applications introduced practical challenges that are different from solving isolated programming exercises. Managing communication between frontend and backend components, handling application data, implementing authentication, and maintaining responsive interfaces require attention to both technical details and the overall application flow.

- Problem-solving: Breaking application requirements into smaller implementable features.
- Debugging: Identifying issues in frontend behavior, API communication and data handling.
- Database thinking: Understanding relationships between users and application entities.
- API understanding: Learning how client-side components communicate with server-side logic.
- User-centric development: Considering usability and responsive design.
- Project management: Completing multiple assignments with different functional requirements.

3.12 Conclusion

The Full Stack Development internship at Thiranex provided practical exposure to the development of complete web applications. Through the portfolio, task management, e-commerce and blog platform assignments, I worked across the major areas of full-stack development, including frontend interfaces, backend APIs, databases, authentication, CRUD operations and responsive design.

The internship helped bridge the gap between academic programming knowledge and practical application development. It also improved my understanding of how different software components interact within a complete web application. The experience will provide a useful foundation for future academic projects, internships and software-development work.

4. REFERENCES

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2. Thiranex, “Certificate of Achievement,” confirming successful completion of the Full Stack Development internship, 2026.
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5. Node.js Documentation, documentation and reference material.
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